

Gold-star manuscript figure

comparison

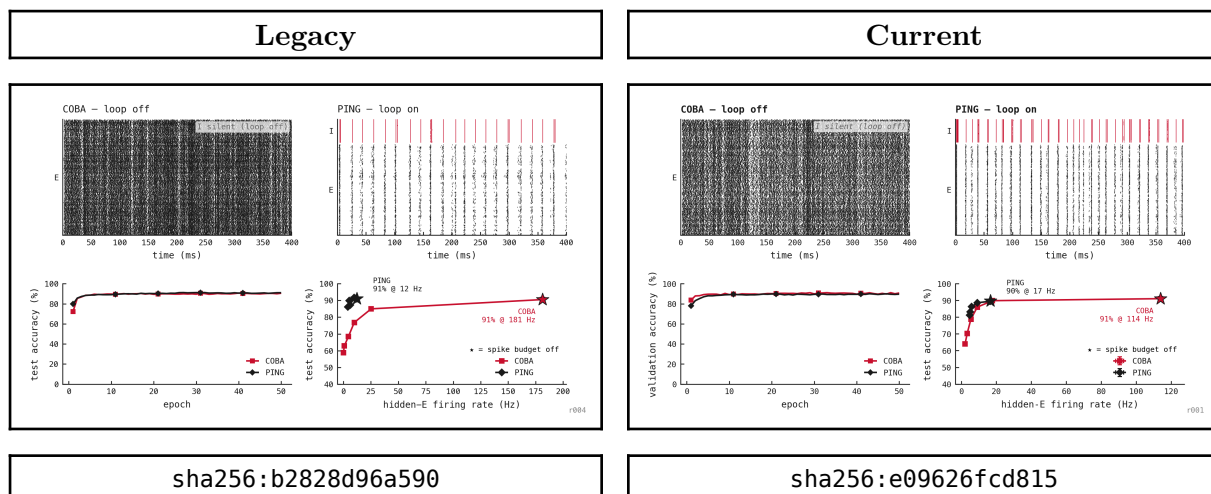
exp093 · 22 August 2026 · Draft

This review compares the manuscript plots retained in both immutable gold-star campaigns. The left column is the historical `legacy-exp022-cc36be1` archive; the right column is the current `ggs-production-composite-20260821-6d9c38eb` publication campaign. A *changed* label means the rendered file bytes differ; it is not by itself a scientific judgement.

Legacy archive	r2://pinglab/campaigns/legacy/exp022-cc36be1
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Trained PING versus COBA

exp025 · results_compound.png · changed

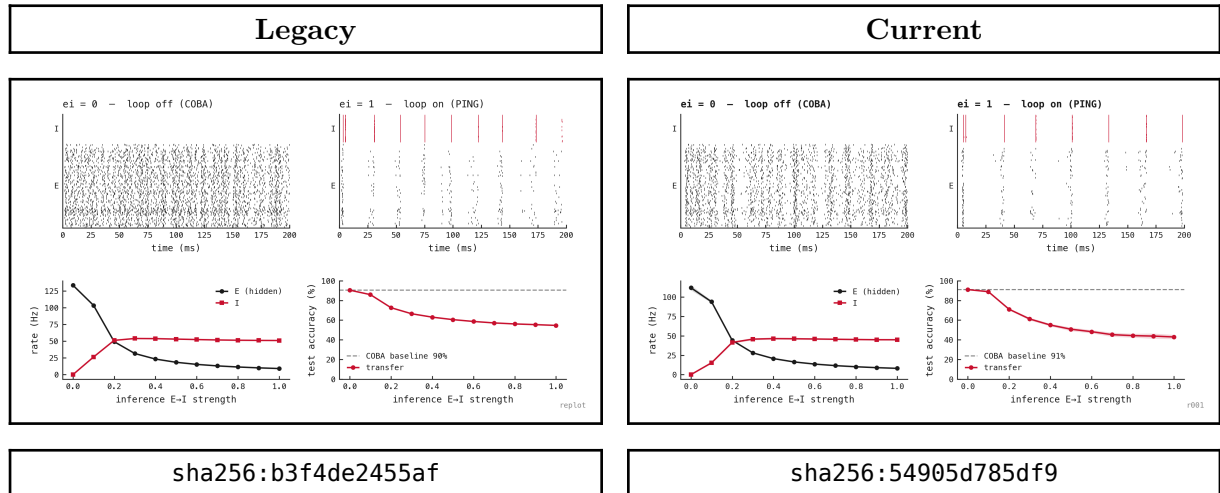


Trained PING matches COBA classification accuracy while operating at an order-of-magnitude lower excitatory firing rate. Both configurations were trained on MNIST by surrogate-gradient descent (§5.4). Top: representative single-trial rasters of the trained networks: COBA fires densely and asynchronously with the inhibitory population silent, whereas PING fires in gamma bands with synchronous inhibitory bursts (red) above excitatory spikes (black). For visualisation, each raster is an extended 400 ms replay of one digit, twice the 200 ms presentation used for training and quantitative evaluation. Bottom left: test accuracy per epoch, both configurations converging to $\approx 90\%$. Bottom right: accuracy-rate frontier traced by sweeping the sample-wise hidden-E rate ceiling (§5.4); each

marker shows mean hidden-E firing rate (abscissa) and test accuracy (ordinate) across three independent seeds, with standard-error bars. PING lies above and to the left of COBA across the sweep. At the unpenalised operating point (starred), PING reaches $\approx 90\%$ at ≈ 16.6 Hz, against COBA's $\approx 91\%$ at ≈ 114 Hz. Source: exp025 rate-attractor analysis in exp024.

Inference-time loop activation

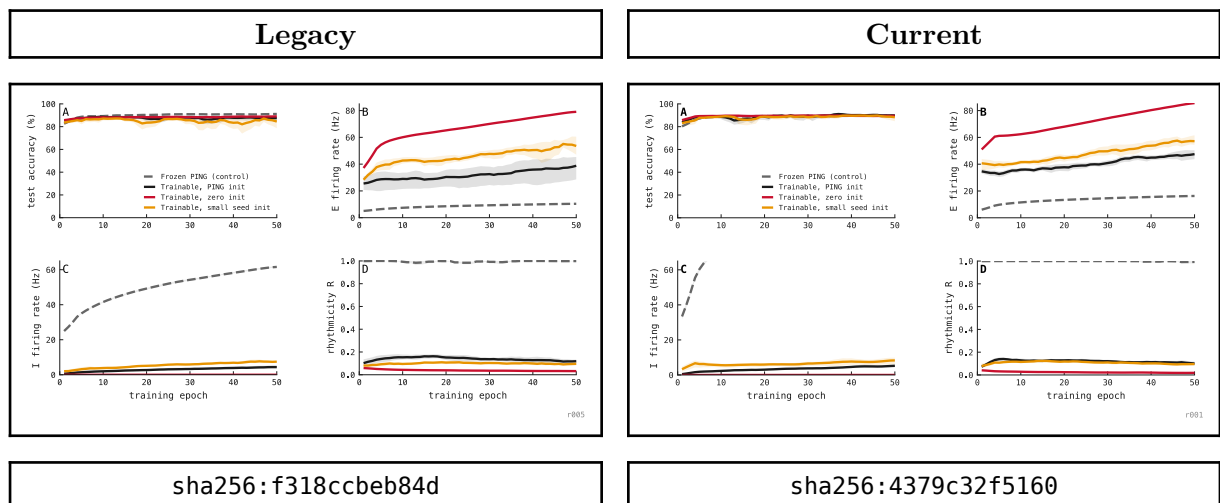
exp038 · loop_transfer_compound.png · changed



Engaging the recurrent loop at inference reproduces the PING firing-rate reduction with no weight update. COBA networks trained with seeds 42–44 are evaluated with recurrent E→I coupling scaled from $ei = 0$ to $ei = 1$, without retraining. Top: illustrative seed-42 rasters at loop-off and full strength. Bottom left: across-seed mean excitatory (black) and inhibitory (red) rates; excitation falls approximately 14-fold (from ≈ 112 to ≈ 8 Hz) as inhibition rises to ≈ 45 Hz. Bottom right: mean accuracy falls from $\approx 91\%$ to $\approx 43\%$ (a ≈ 48 pp cost). Shaded bands show sample SD across seeds. The rate gating appears when the loop is wired in, whereas the accuracy loss reflects the absence of a feedforward compensation learned in the presence of the loop. Source: exp038.

Released-loop training

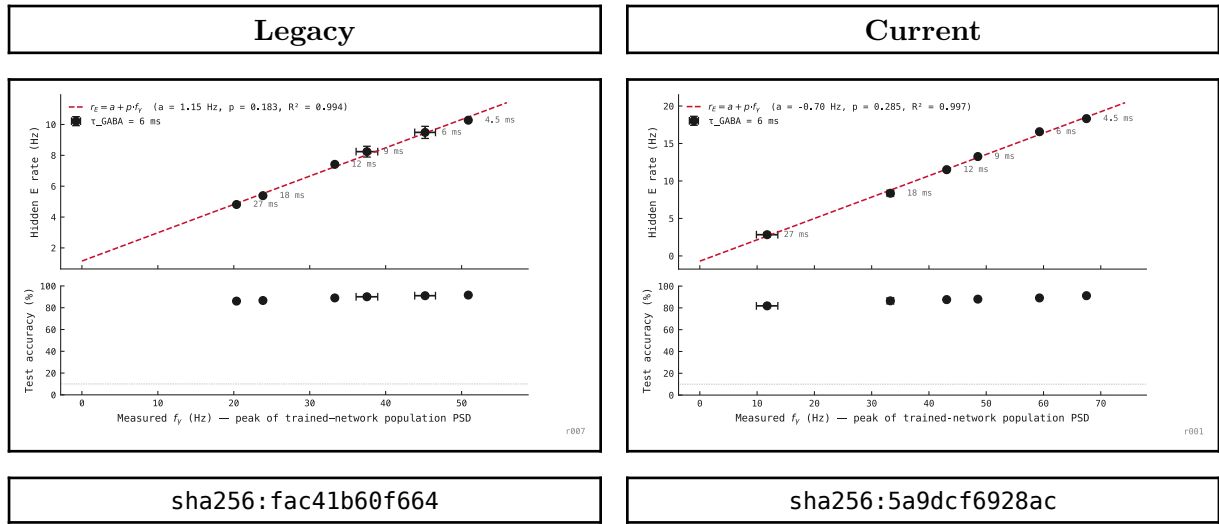
exp049 · training_curves.svg · changed



Releasing the loop weights to gradient descent prunes the rhythm within a single epoch, from every initialisation. Per-epoch training metrics with the recurrent weights W^{EI}, W^{IE} made trainable under the Dale’s-law clamp, over 50 epochs on MNIST. Lines are the mean of three seeds (42–44); shading is the across-seed range. Conditions differ only in initialisation of the loop weights: canonical PING values (black), zero (red), and $0.1 \times$ canonical (amber), against a frozen-PING control (grey, dashed). **(A)** Test accuracy: all conditions overlap at ≈ 84 – 91% . **(B)** Mean excitatory firing rate: the trainable conditions rise to ≈ 49 – 88 Hz as the loop is dismantled, while the frozen control remains gated near 17 Hz. **(C)** Mean inhibitory firing rate: the trainable conditions collapse to ≈ 0 Hz within a few epochs, while the frozen control’s inhibitory rate rises to ≈ 108 Hz as its readout trains. **(D)** Lobe–trough rhythmicity contrast: the frozen control holds at ≈ 1 while every trainable initialisation drains to ≈ 0.02 – 0.11 . Source: exp049.

Rate versus gamma frequency

exp041 · rate_vs_fgamma.svg · changed

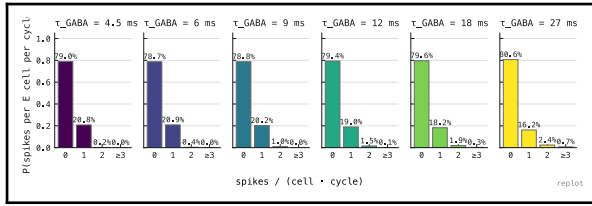


Post-training excitatory rate covaries affinely with gamma frequency across a sweep of inhibitory decay kinetics. Networks were trained from scratch at each of six values of the GABA decay constant $\tau_{\text{GABA}} \in \{4.5, 6, 9, 12, 18, 27\}$ ms. Changing τ_{GABA} alters both the realised gamma frequency and the duration and integrated influence of inhibition (§5.4). Top: mean post-training excitatory firing rate against measured f_γ ; markers are per-condition means over three seeds, with error bars ($\pm$ SD) on both axes. The linear fit $r_E = -0.7 + 0.285f_\gamma$ passes through every error bar ($R^2 = 0.997$). Bottom: mean test accuracy declines from 91.2% at $\tau_{\text{GABA}} = 4.5$ ms to 81.9% at 27 ms, a 9.3 percentage-point tradeoff. The sweep establishes covariance with realised frequency, not an independent causal effect of frequency alone. Source: exp041.

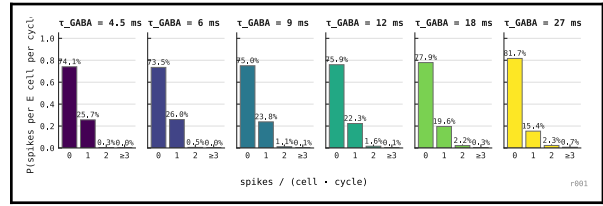
Spikes per gamma cycle

exp046 · spikes_per_cycle_distribution.svg · changed





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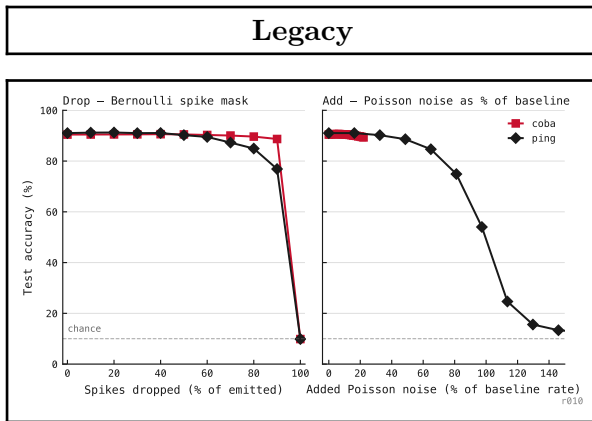


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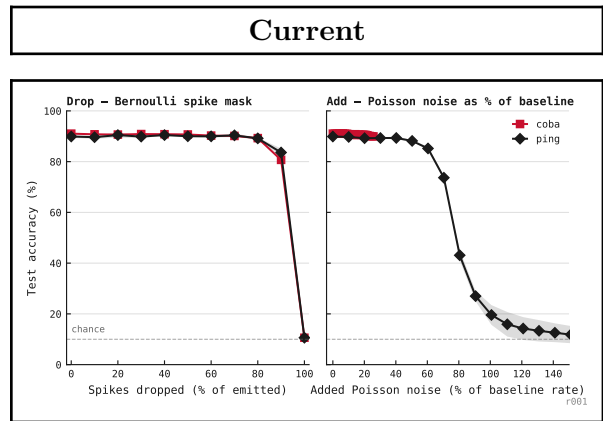
The affine rate law follows from a near-binary per-cycle firing statistic: each excitatory cell contributes at most one spike per gamma cycle. Excitatory spikes were resolved into (cell, cycle) pairs by assigning each spike to the gamma cycle inferred from peaks of the population inhibitory-burst rate (§5.5). Each panel shows the distribution of spikes per pair (0, 1, 2, or ≥ 3) at one value of τ_{GABA} . Across the sweep, $P(0) \approx 75\%$, $P(1) \approx 24\%$, and the multi-spike fraction (≥ 2) stays near $\approx 1\%$; pooled over all conditions, 98.85% of pairs contain at most one spike. Source: exp046.

Spike perturbation robustness

exp037 · perturbation_curves.svg · changed



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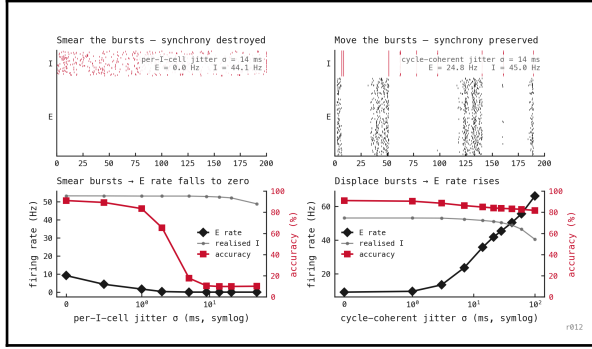
The gating is robust to spike deletion but fragile to spike addition, the expected signature of a phase-based code. PING and COBA networks trained with seeds 42–44 are perturbed at inference without retraining. Left: deletion of a random fraction of emitted spikes. PING retains $\approx 89\%$ mean accuracy through 80% deletion, close to its unperturbed value. Right: addition of Poisson spikes, expressed relative to each population's baseline rate. PING falls toward chance as off-phase spikes disrupt the rhythm; COBA, having no rhythm to protect, is more tolerant to addition. Lines show across-seed means and shaded bands show sample SD. Source: exp037.

Inhibitory timing perturbations

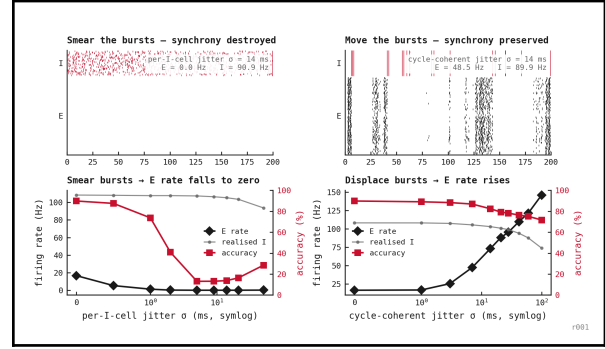
exp042 · rhythm_compound.png · changed

Legacy

Current



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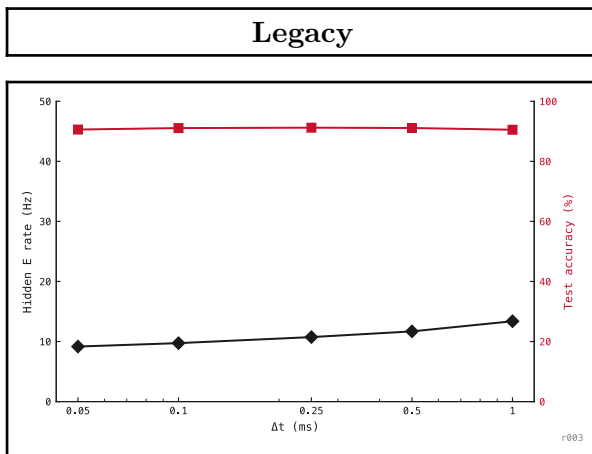


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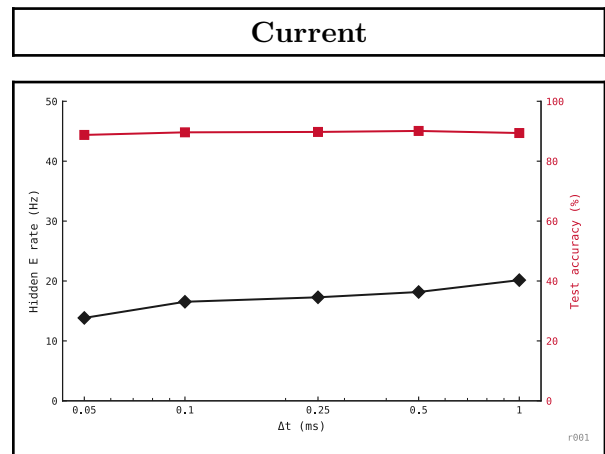
Two inhibitory-jitter manipulations at the same magnitude that hold the mean inhibitory rate fixed drive the excitatory rate in opposite directions, isolating timing from level. The trained PING inhibitory spike train is perturbed at inference while the mean per-cell inhibitory rate is held constant. Both arms use the same jitter magnitude, $\sigma = 14$ ms — only the *kind* of jitter differs. Top: single-trial rasters (E black, I red). Bottom: mean excitatory rate (black) and accuracy (red) versus jitter magnitude σ , with the realised mean inhibitory rate overlaid (grey). Left: per-cell jitter smears each burst into a continuous shunt; the excitatory rate falls to ≈ 0 Hz (inhibitory rate ≈ 105 Hz) and accuracy falls to chance. Right: cycle-coherent jitter displaces whole bursts while preserving within-burst synchrony; the excitatory rate rises from ≈ 17 to ≈ 73 Hz (inhibitory rate ≈ 103 Hz, matched to the left arm) and accuracy holds high. Both arms are read where the realised inhibitory rate is matched to within a few percent; at larger σ the cycle-coherent excitatory rate climbs further, but the finite trial window truncates the most-displaced bursts and realised inhibition falls, so the strict comparison is anchored here. Identical mean inhibition, opposite excitatory outcome: the gate is the phase structure of inhibition, not its level. Source: exp042.

Integration-timestep invariance

exp044 · dt_sweep.svg · changed



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The firing-rate reduction is a physical-time property, invariant to the integration timestep over a twentyfold range. The network was trained and evaluated at matched integration timestep Δt for each value across $[0.05, 1.0]$ ms (logarithmic abscissa). Left

ordinate (black diamonds): post-training mean excitatory rate, confined to a 13.6–20.2 Hz band and non-monotonic in Δt , so finer stepping does not buy a lower rate. Right ordinate (red squares): test accuracy, flat within 2.2 pp (88.2–90.4%). Because both training and inference use the same Δt at each point, the figure tests invariance of the training-plus-inference pipeline, not the generalisation of one trained network to a varied inference step. Source: exp044.